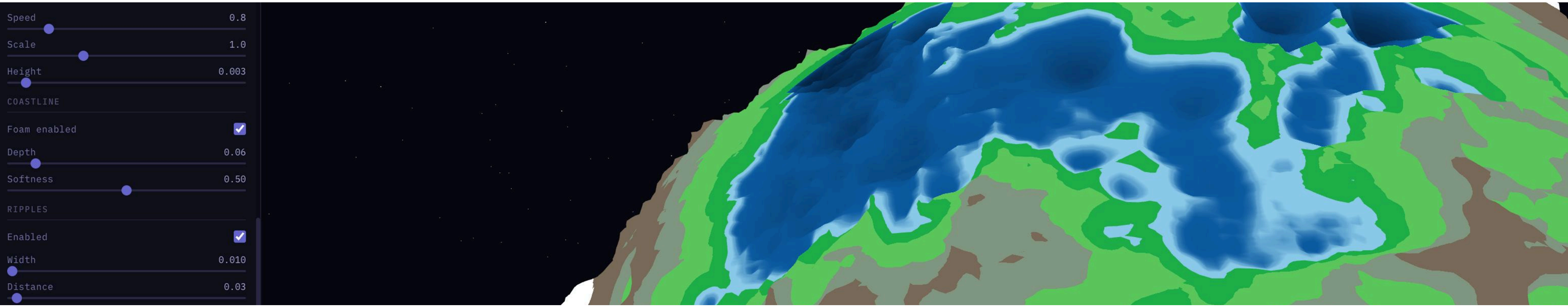
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Class 04

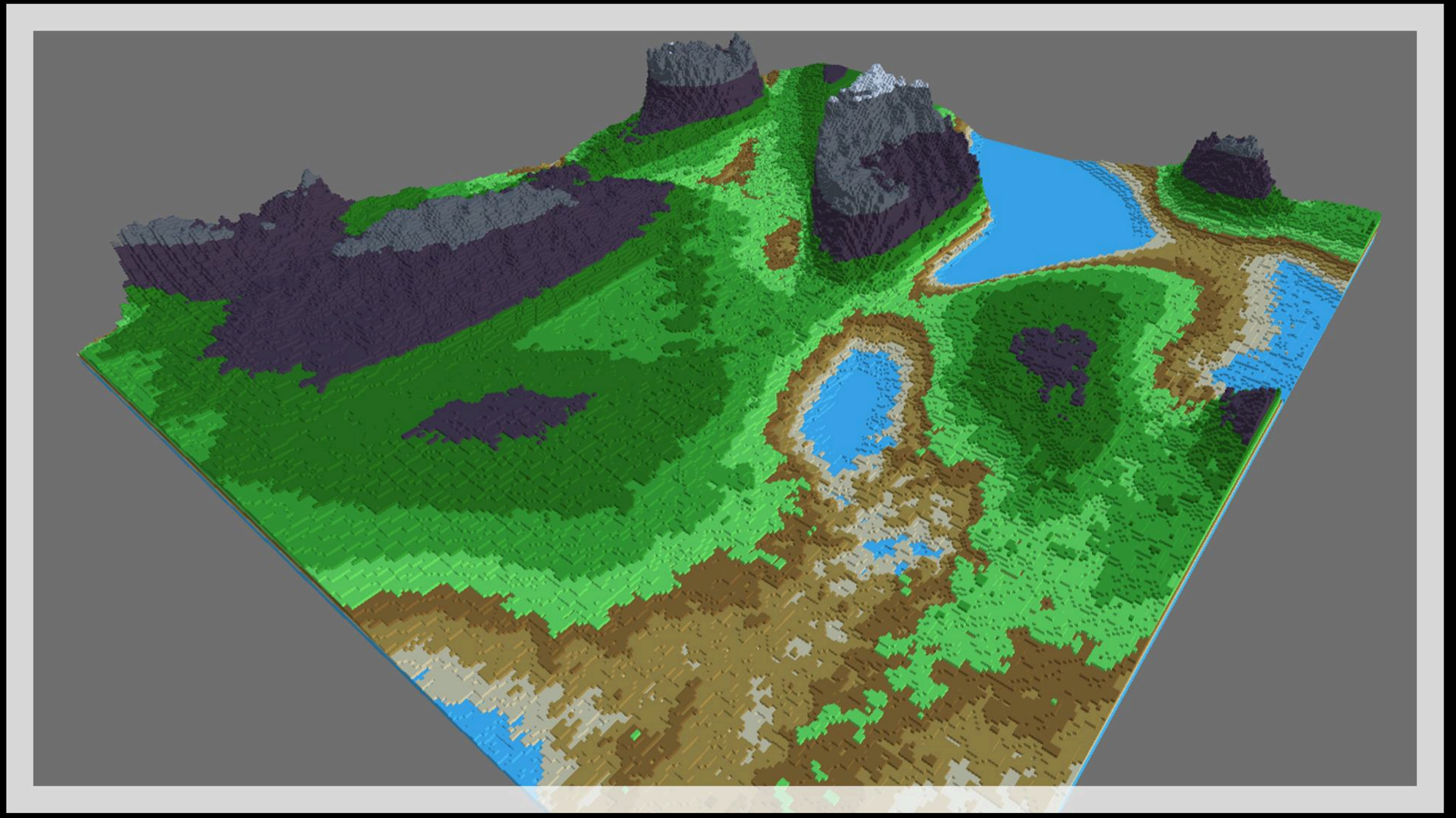
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Procedural Generation Techniques

PCG 02: Voxels

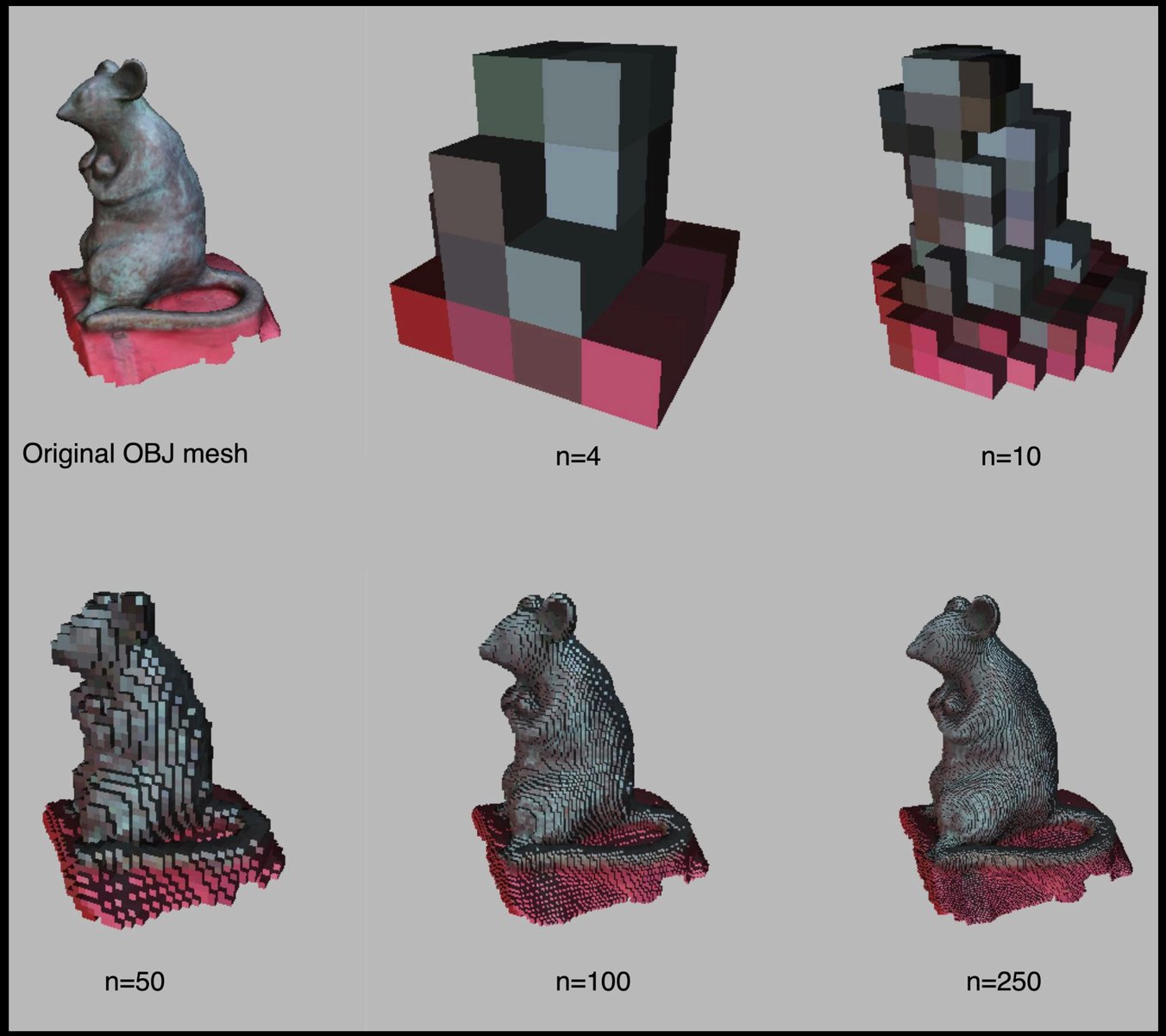
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01 - Voxels



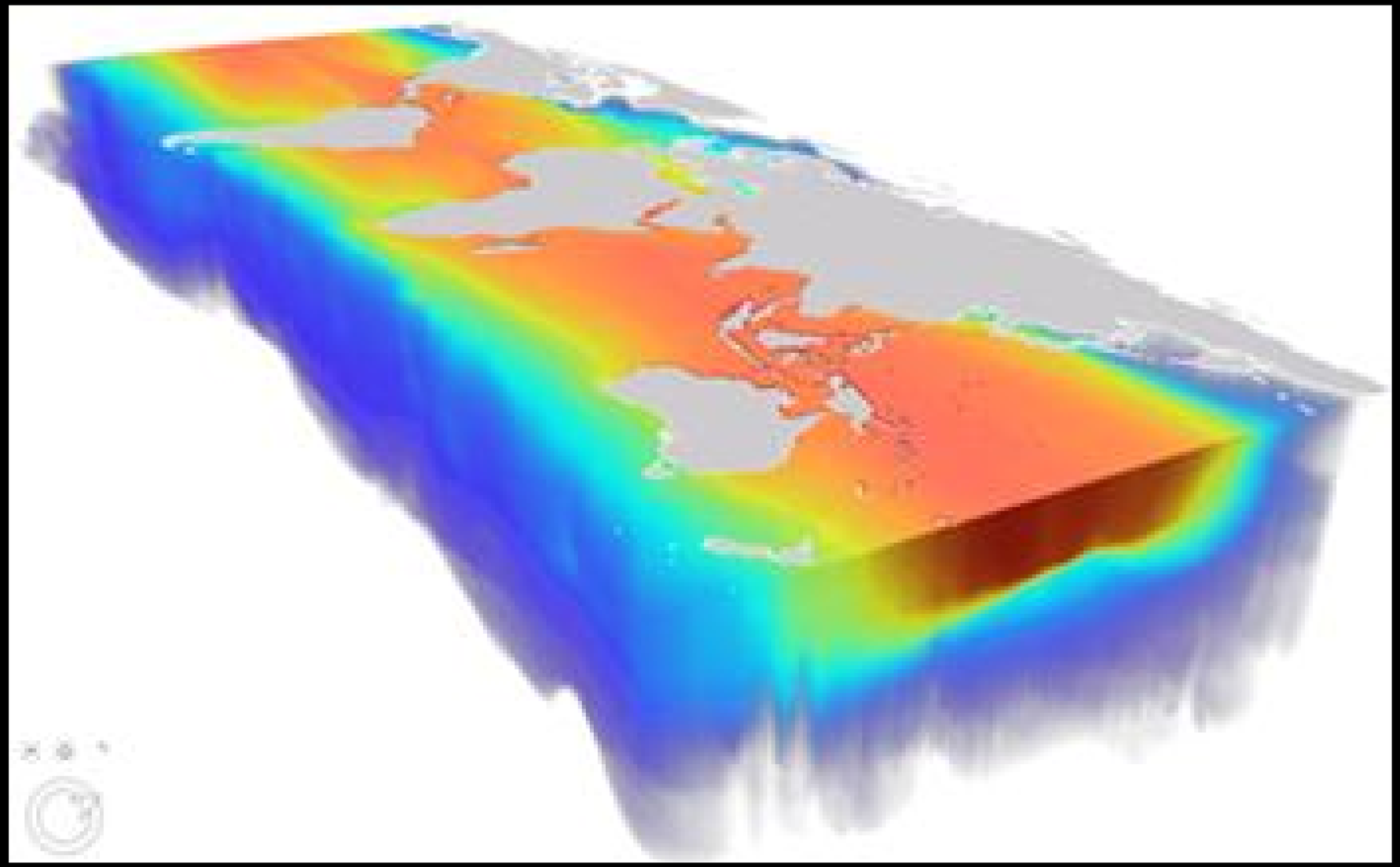
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01 - Voxels
Resolution



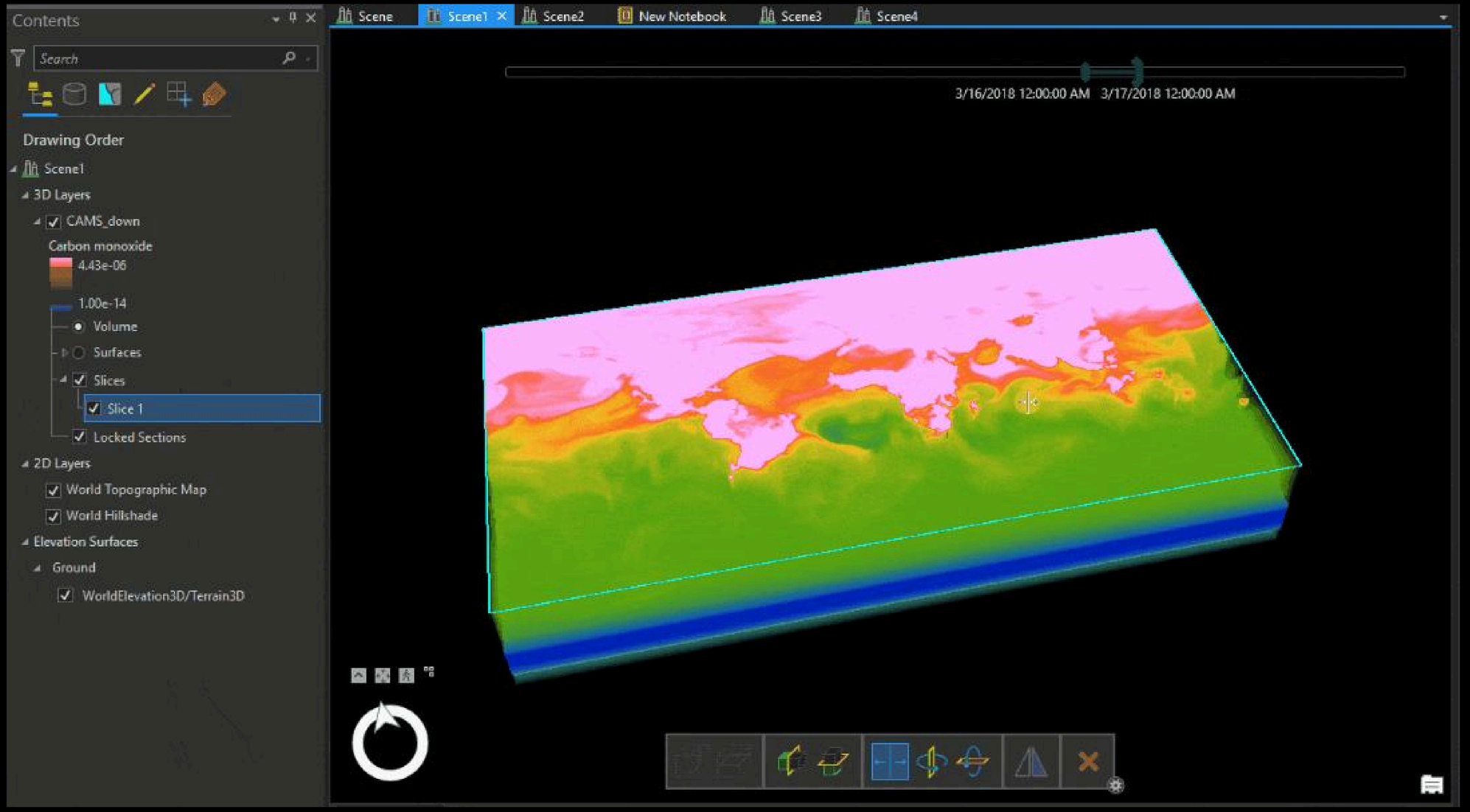
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01 - Voxels



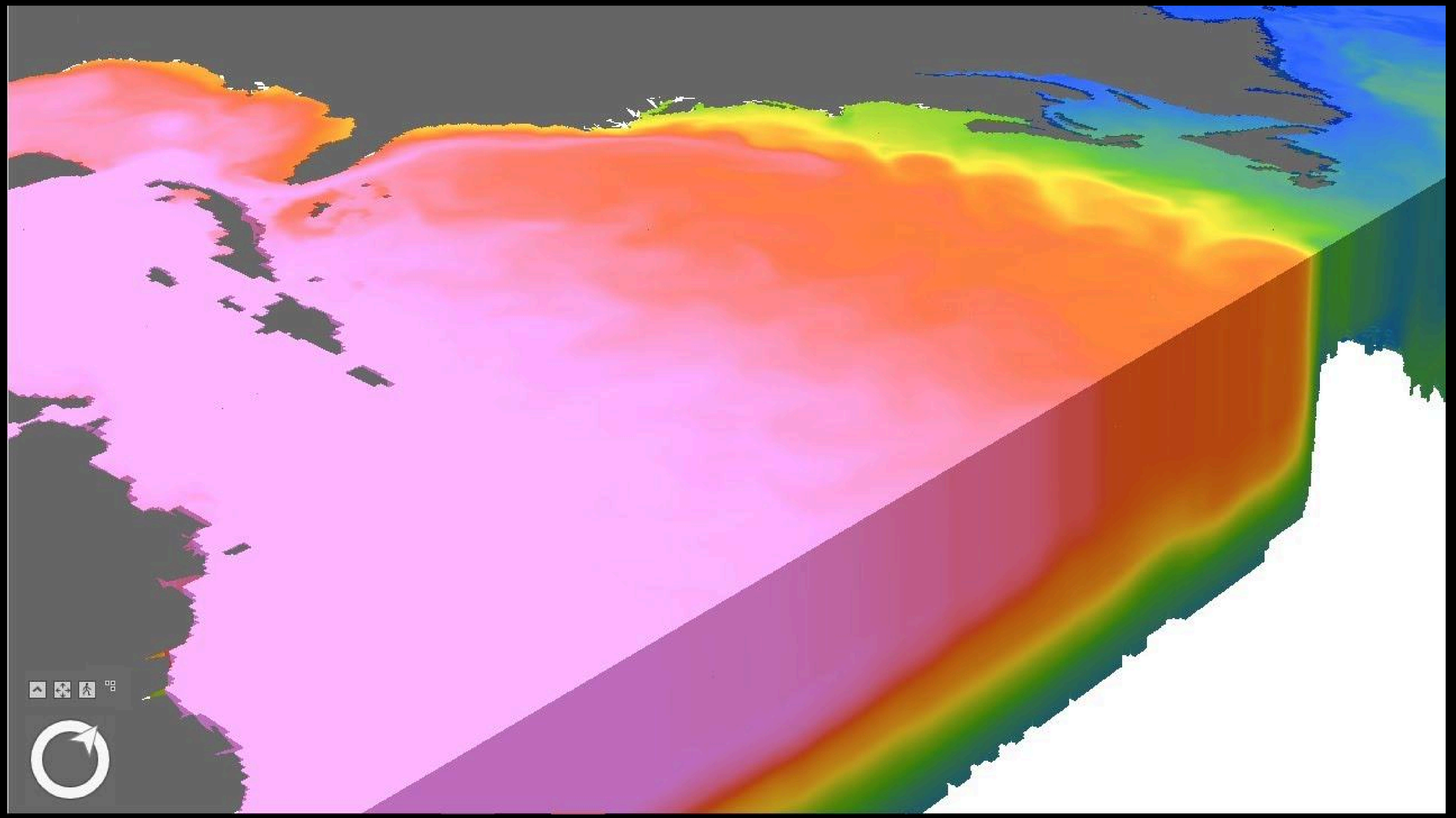
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01 - Voxels



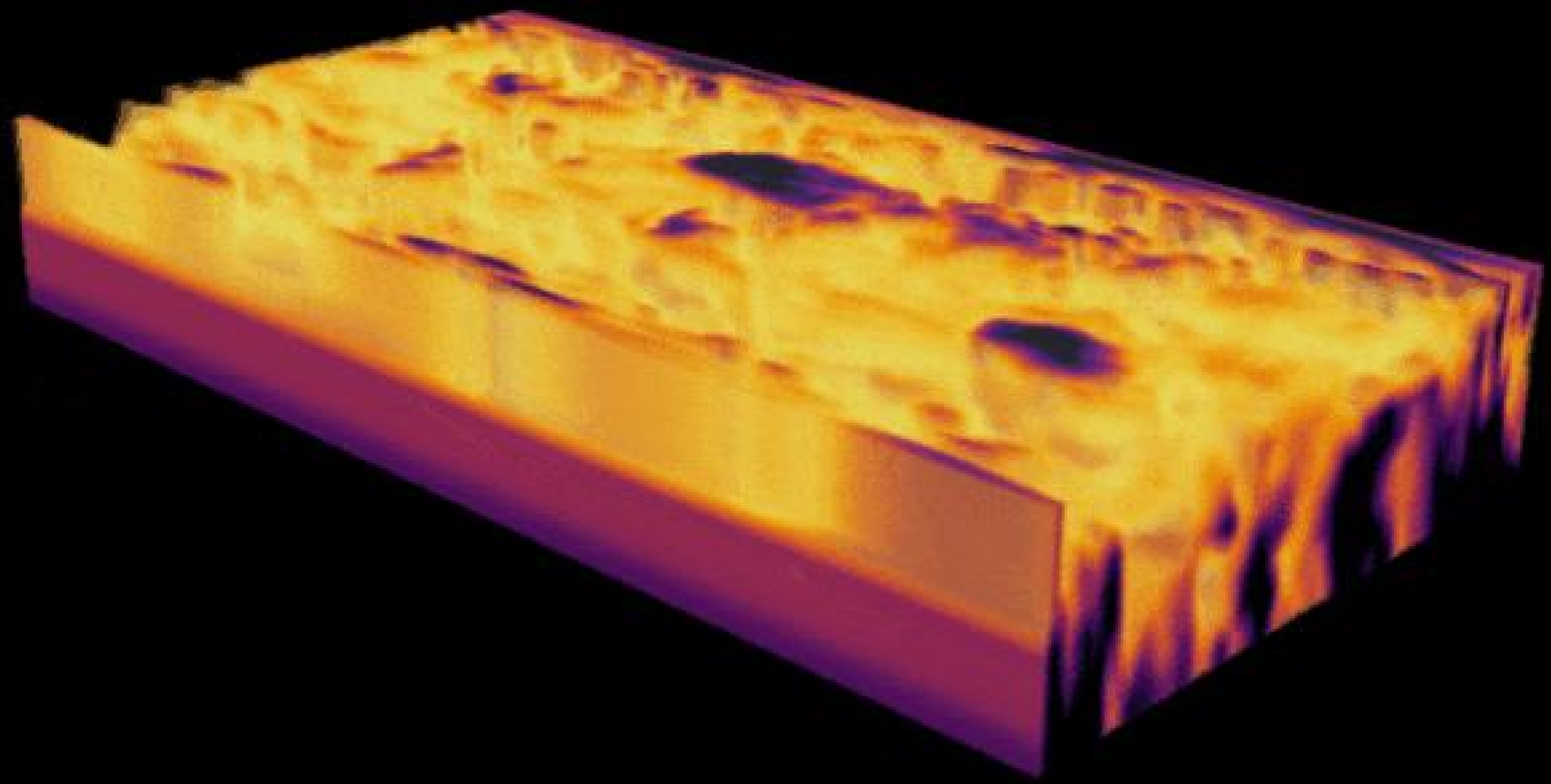
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01 - Voxels



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01 - Voxels



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给空间中的每个位置一个数，再根据这个数判断形状在哪里。

01 - Voxels
Volume / Density Shapes

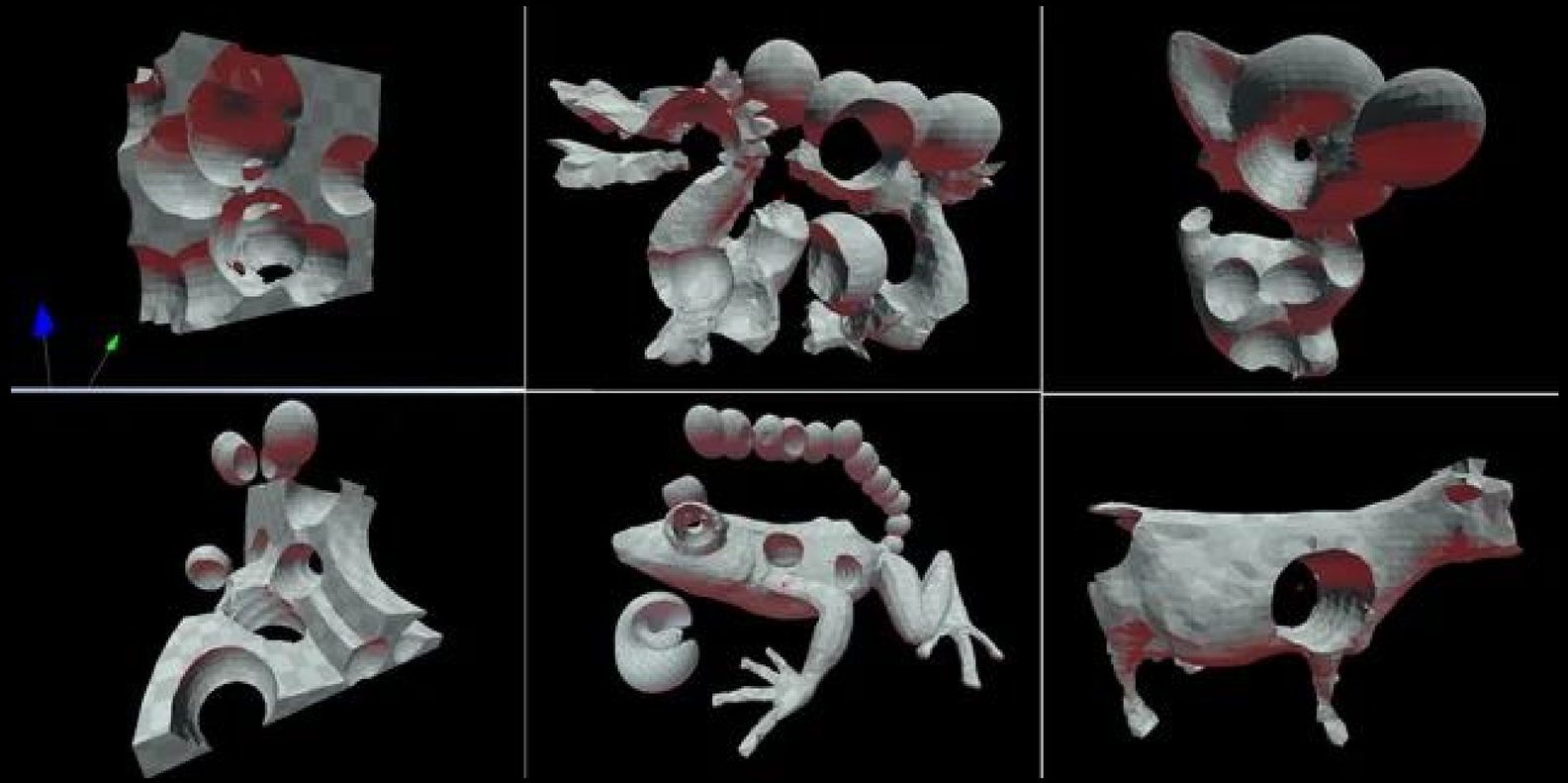
高度函数可以造普通地形；
三维噪声可以造不规则实体；
分层函数可以造阶梯地貌；
球形距离加上变化可以造星球。

Shape	Density term	Gives
Ground plane	`h(x,z) - y`	heightfield terrain
3D fBm	`fbm3d(p)`	blobby, sponge-like mass
Ridged 3D	`1 - |fbm3d(p)|`	sheets and walls
Terraced	`ground` with `y` replaced by `floor(y/k)*k + smooth part`	mesas, stepped cliffs
Floating islands	`fbm3d(p) - falloff(y)` where falloff is small in a height band	sky islands
Planet	`radius + h(normalize(p)) - length(p)`	the Class 01 §6.4 planet, now diggable
Strata	`ground + sin(y * k + fbm2d(x,z))`	layered sedimentary bands

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01 - Voxels
CSG - Constructive Solid Geometry

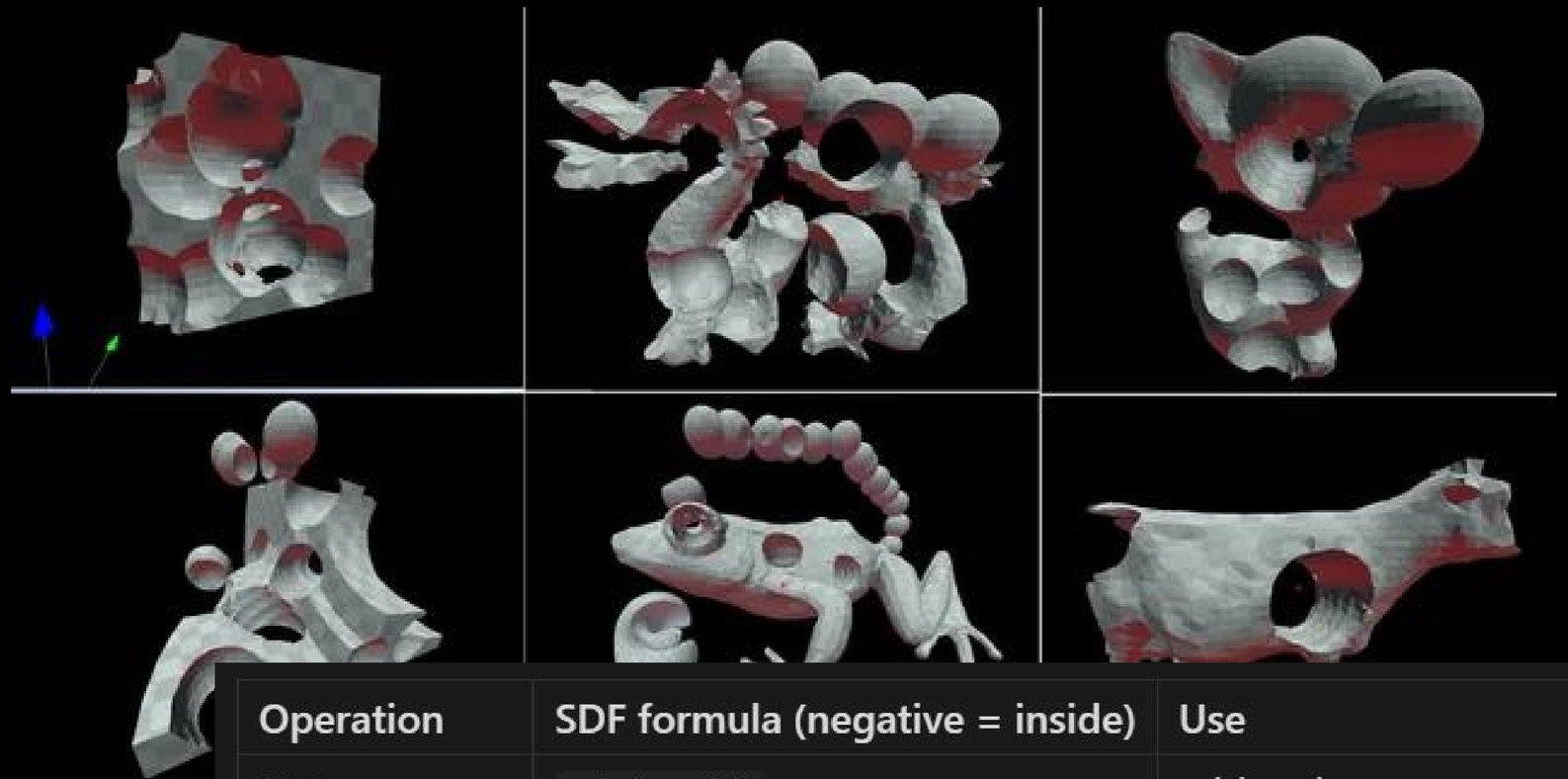
构造实体几何



把简单形状依次组合，就能生成复杂形状。比如先造一块岩石，再挖出一个洞。

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01 - Voxels
CSG -
Constructive Solid Geometry



合并

取共同部分

从一个形状里减去另一个

让连接处更圆滑

形成薄壳

Operation	SDF formula (negative = inside)	Use
Union	`min(a, b)`	add rock
Intersection	`max(a, b)`	clip to a region
Subtraction	`max(a, -b)`	carve a tunnel or room
Smooth union	`smin(a, b, k)`	organic joins, no crease
Shell	`abs(a) - thickness`	hollow objects, walls

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01 - Voxels
CSG -
Constructive Solid Geometry



洞穴或隧道的生成方法

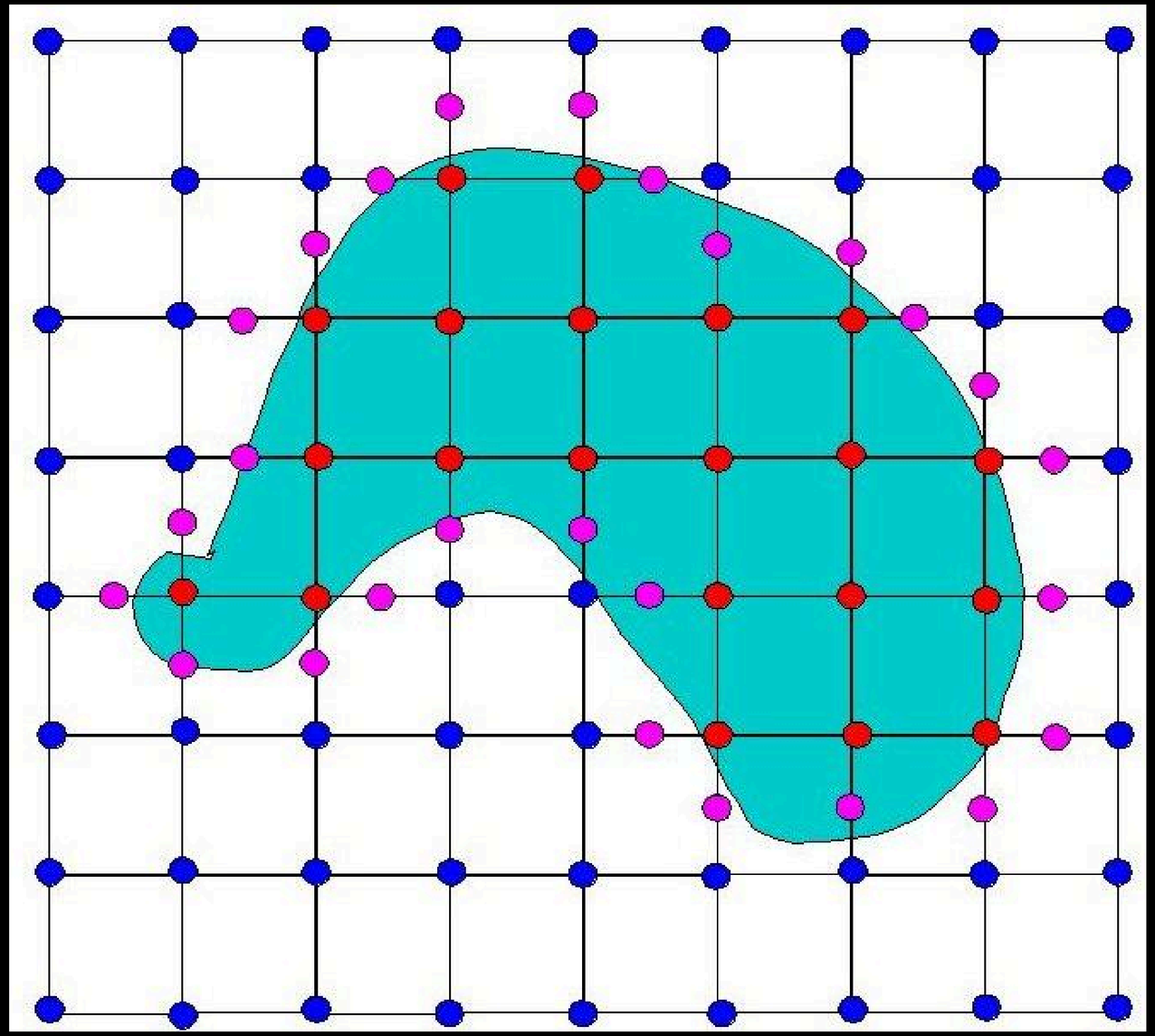
Method	Look	Evaluative?	Cost	Control
Threshold	chambers	yes	one noise per voxel	density via threshold
Intersection	tunnels	yes	two noises per voxel	thickness via epsilon
Worms	tunnels with intent	no	per-worm, cross-chunk lookup	high: length, radius, direction
3D CA	organic pockets	no	per step × 26 neighbours	low (see below)

表格比较了阈值、形状相交、沿路径挖隧道等方法。
有的容易计算但难控制洞的走向；有的能明确控制路径，但要处理跨越多个 chunk 的情况。

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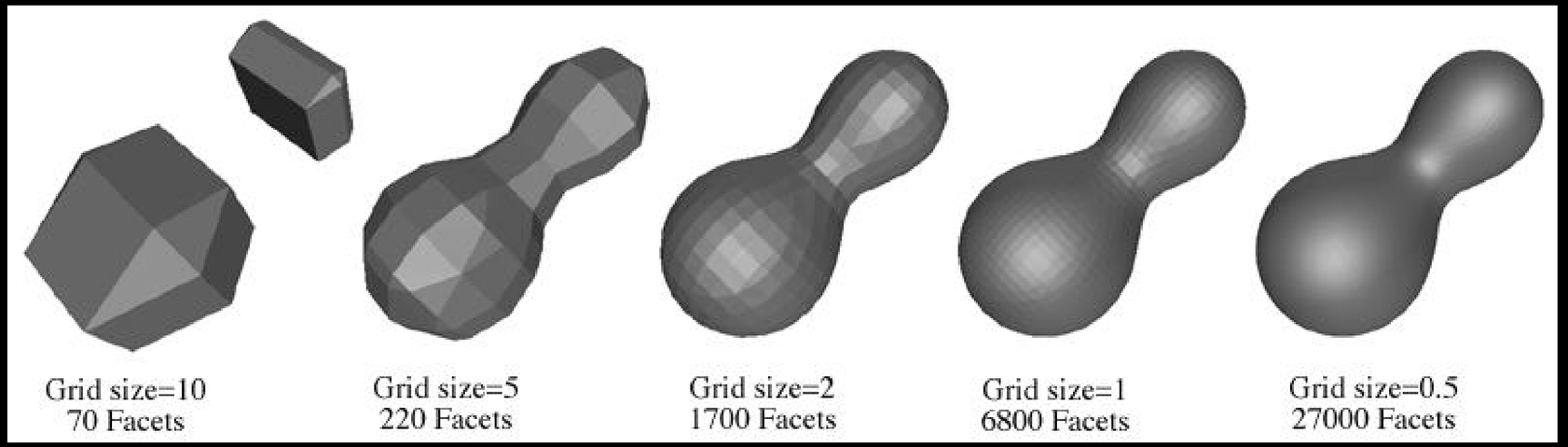
01 - Voxels
Mesh Generation
Marching cubes

用二维网格帮助理解：
一个连续形状穿过格子时，检查格子角上的数值，并在格子内部找出边界。
三维的 Marching Cubes 用类似方法，在小立方体中生成表面三角形。



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网格越细，物体越圆滑，但面数迅速增加。



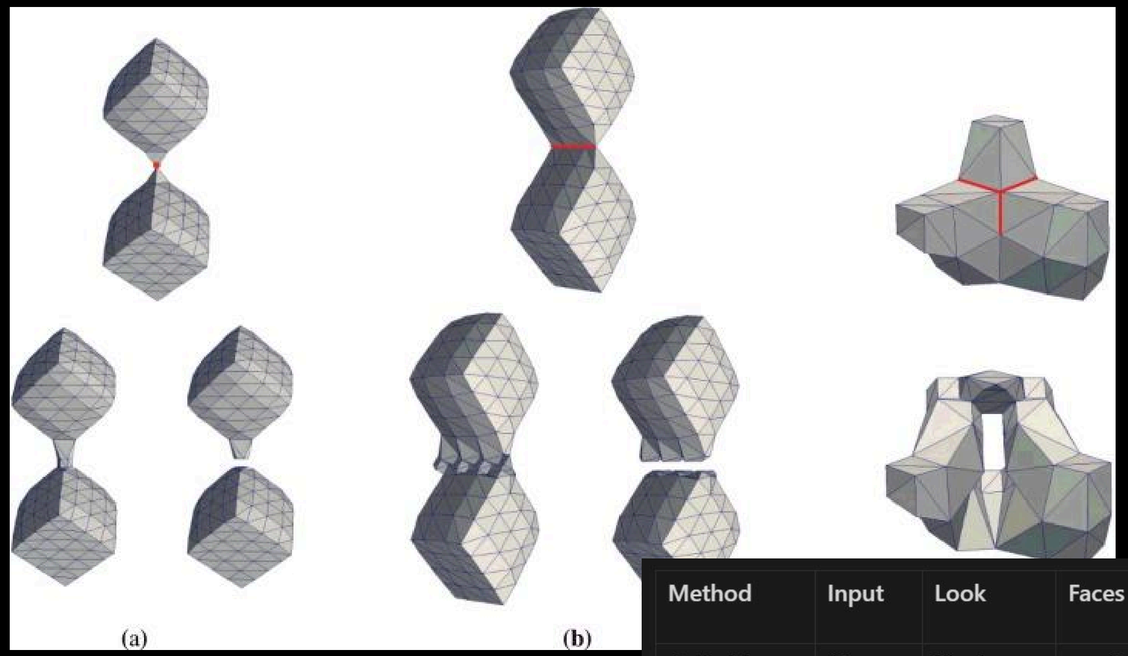
不同网格生成方法的比较

01 - Voxels
Mesh Generation
Marching cubes

下方表格比较直接显示方块面、Greedy Meshing、Marching Cubes 等方法。
体素是保存形状的数据；mesh 是拿来渲染表面的几何面片，两者不要混为一谈。

Method	Input	Look	Faces	Sharp features	Topology	Relative cost
Culled faces	binary	blocky	medium	yes (axis-aligned only)	watertight, often non-manifold	lowest
Greedy	binary	blocky	lowest	yes (axis-aligned only)	T-junctions	low–medium
Marching cubes	density	smooth, bevelled	highest (tris)	no	manifold if disambiguated	medium
Marching tetrahedra	density	smooth	very high	no	unambiguous	medium–high
Surface nets	density	smooth	low (quads)	no	mostly manifold	low–medium
Dual contouring	Hermite	smooth + sharp	low (quads)	**yes**	risk of non-manifold/self-intersection	high

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01 - Voxels
Mesh Generation
Dual Contouring

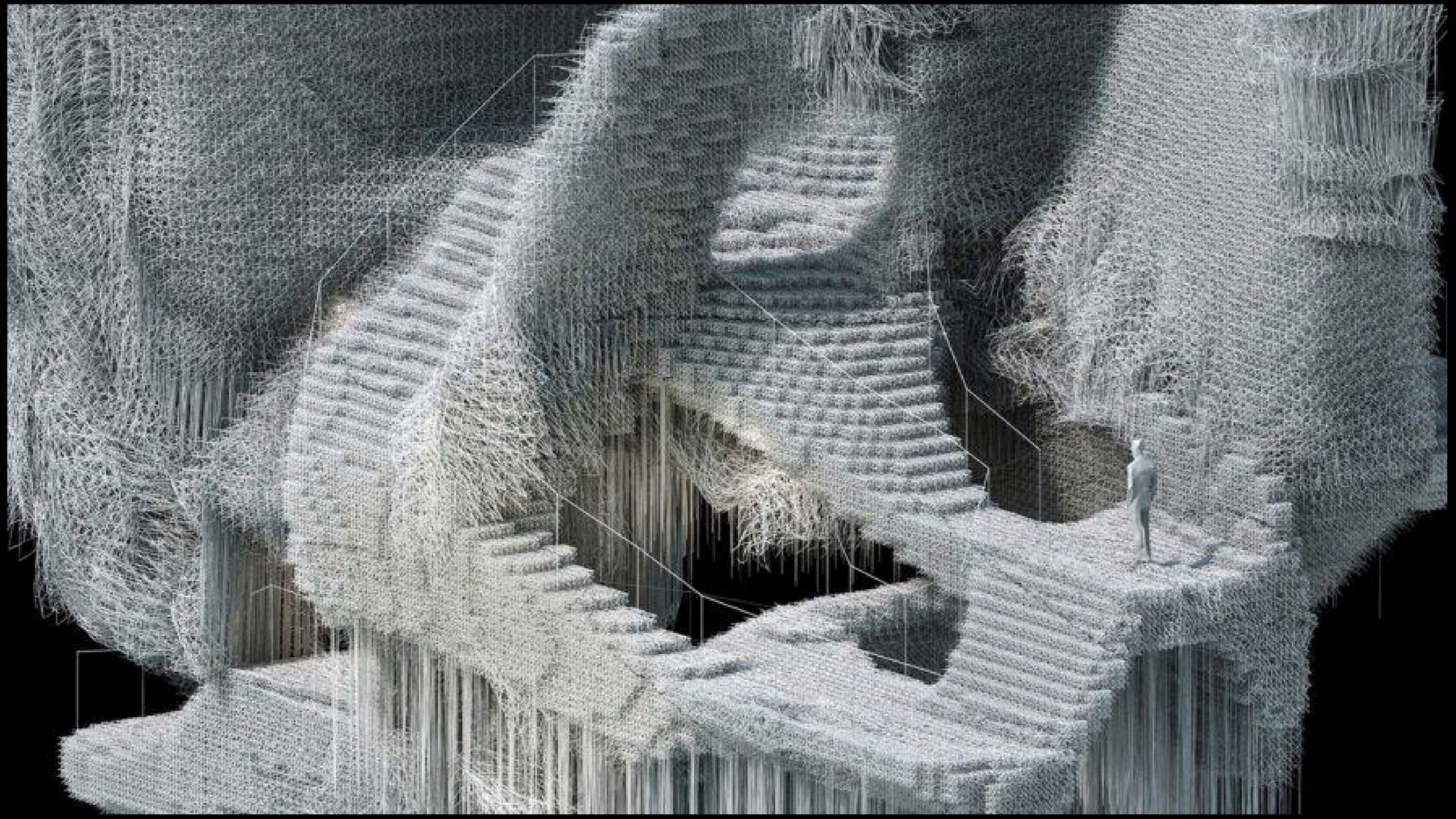
这是另一种从体积数据提取表面的方法。
相比追求圆滑表面的 Marching Cubes，它也适合保留某些棱角；
代价是实现和处理特殊连接情况更复杂。

Method	Input	Look	Faces	Sharp features	Topology	Relative cost
Culled faces	binary	blocky	medium	yes (axis-aligned only)	watertight, often non-manifold	lowest
Greedy	binary	blocky	lowest	yes (axis-aligned only)	T-junctions	low–medium
Marching cubes	density	smooth, bevelled	highest (tris)	no	manifold if disambiguated	medium
Marching tetrahedra	density	smooth	very high	no	unambiguous	medium–high
Surface nets	density	smooth	low (quads)	no	mostly manifold	low–medium
Dual contouring	Hermite	smooth + sharp	low (quads)	**yes**	risk of non-manifold/self-intersection	high

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01 - Voxels
Hi-Res

高分辨率体素



大量细小元素能够表现丰富的空间细节，但数据量和绘制成本也会上升。

Voxel Cloud by Julian Edelmann

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01 - Voxels
Hi-Res



Chunking 是把大世界切成小块，只生成、更新或显示需要的块；
threads 是让部分计算并行进行。这页对应你之前问的性能问题：整个世界若作为一个整体反复重算，会越来越慢。

Voxel Chunking in Godot by MakerTech

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01 - Voxels
Interactive

展示体素世界可以支持大范围地形和玩家交互。



Minecraft videogame

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01 - Voxels
Interactive



展示可改变、可破坏的场景；当环境发生变化时，相关数据和表面需要更新。

Teardown videogame

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01 - Voxels
Interactive



展示玩家能直接改变地形形状。这里很适合用体积数据表示“挖掉或加上一块空间”。

Astroneer videogame

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Assignment 1:

-Create a Voxel Terrain in a different tab
-Explore and implement different density shapes
-Explore and understand CSG techniques - sequential operations
-Consider the limitations of size and performance → explore and document why we need “chunking”
-Implement a “meshing” solution such as marching cubes
-Document and understand the features of alternative meshing techniques
-Explore ways to optimize a voxel structure

你需要在单独的 tab 建立体素地形；
尝试多种密度形状和连续的 CSG 运算；
解释为什么需要 chunking；
实现一种 meshing 方法（如 Marching Cubes）；
比较替代方法，并探索优化。
这里的“document”意味着除了做出来，还要说明选择和限制。

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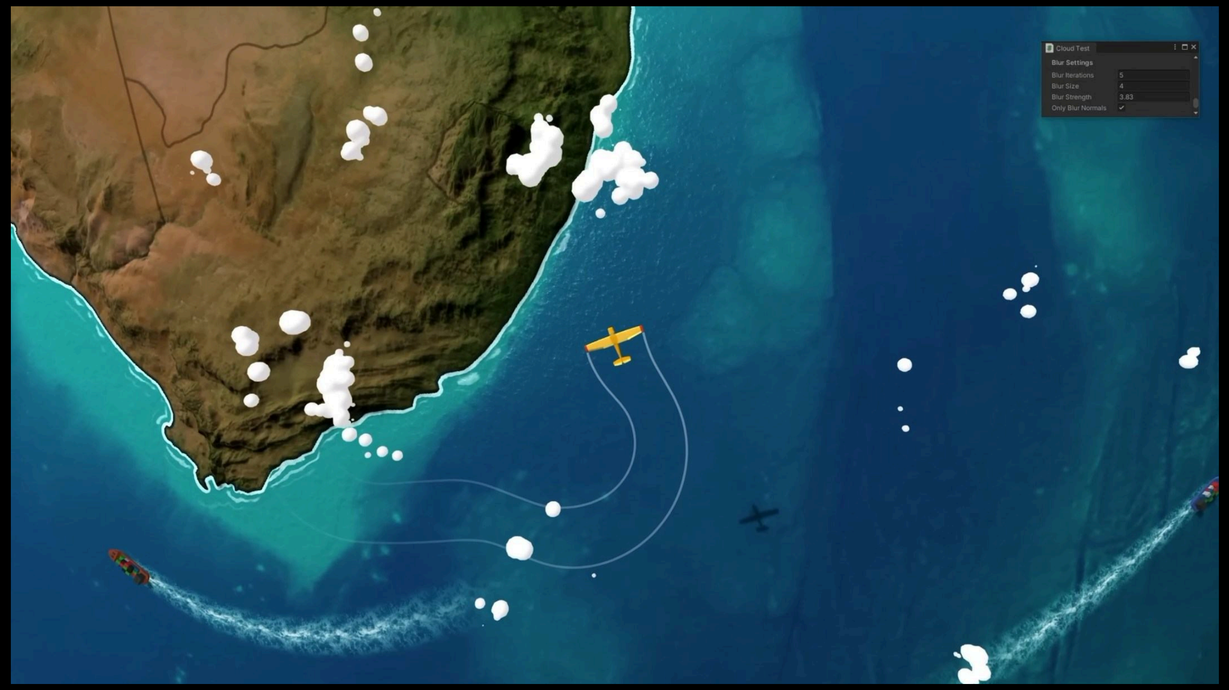
Additional resources:

Subastian Lague:

Geographic data:
https://www.youtube.com/watch?v=sLqXFF8mlEU

Voxels:
https://www.youtube.com/watch?v=M3iI2l0ltbE

Terraforming:
https://www.youtube.com/watch?v=vTMEdHcKgM4



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Procedural Generation Techniques

PCG 02: Backend

用户的数据如何保存和再次读取

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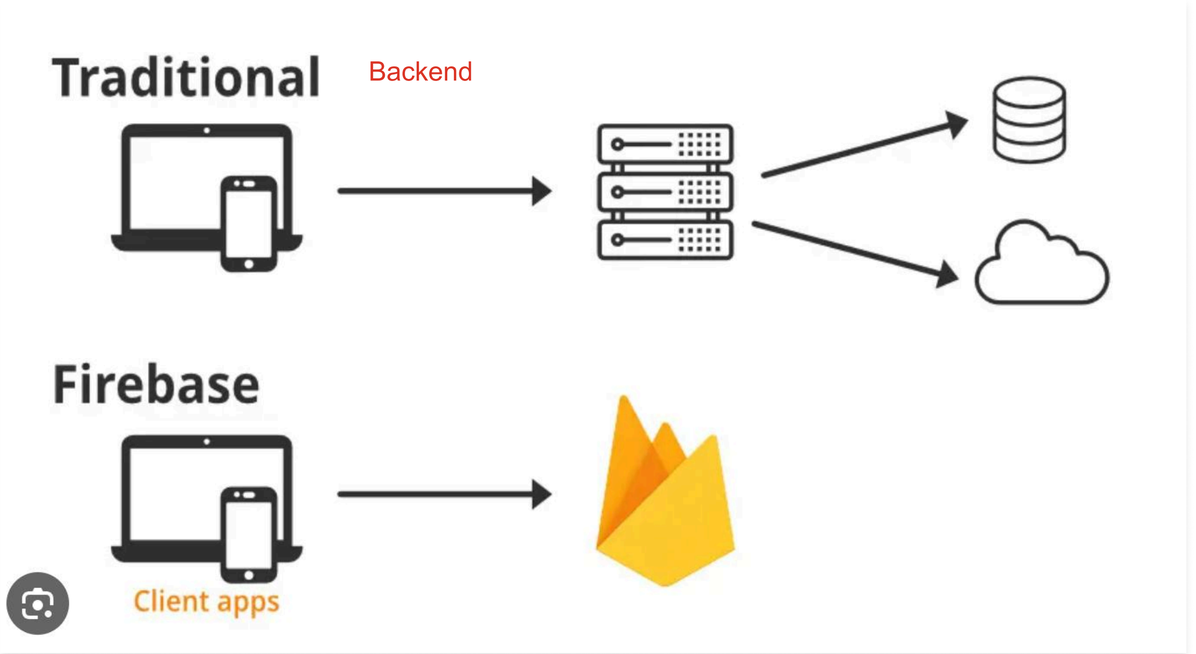
SAAS - Software as Service

通过网络提供的软件服务。

多个产品作例子，带出在线应用通常需要账户与持久化数据。

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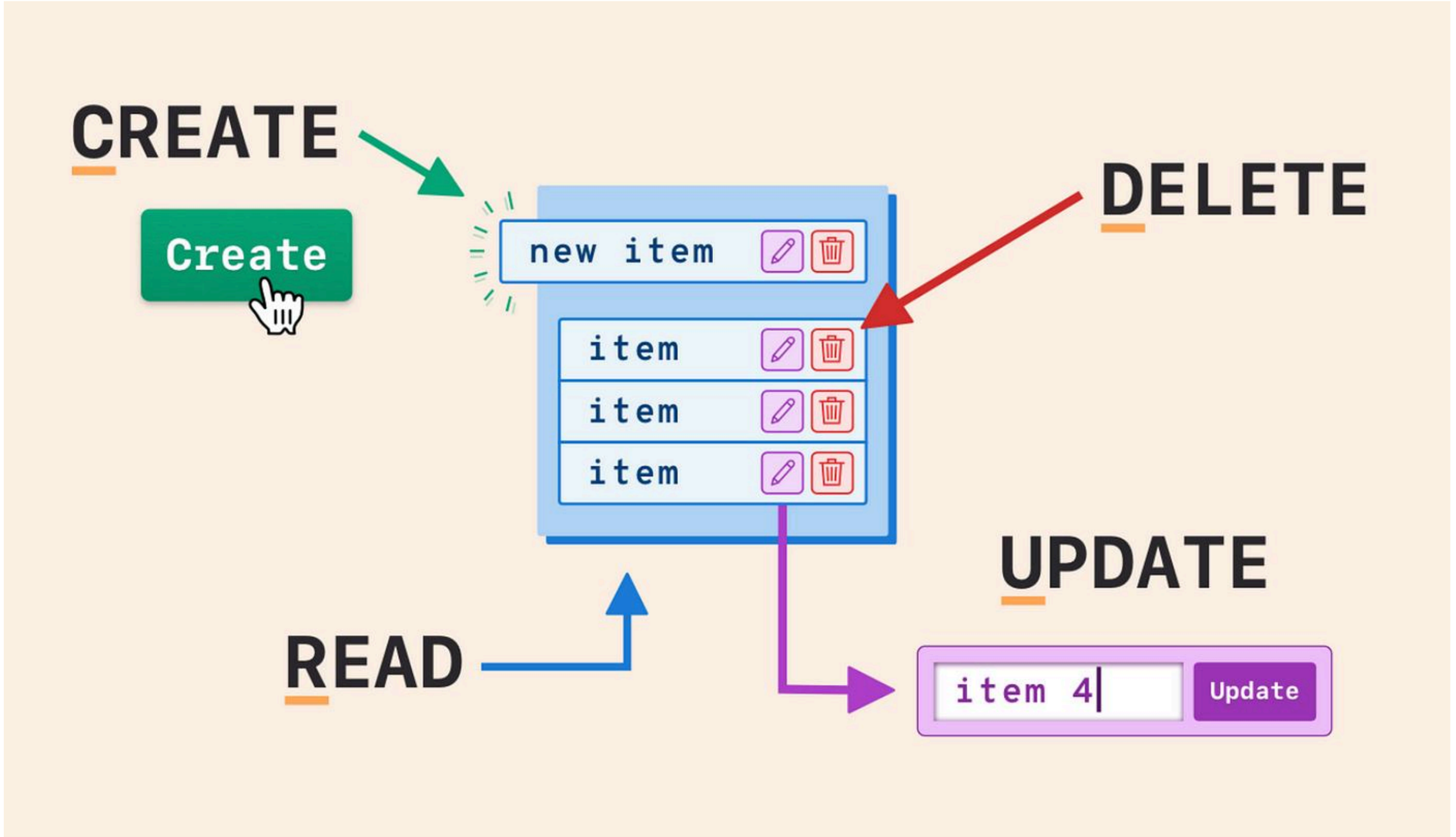




Firebase 承担账户、数据库、文件和网站发布等工作

Backend

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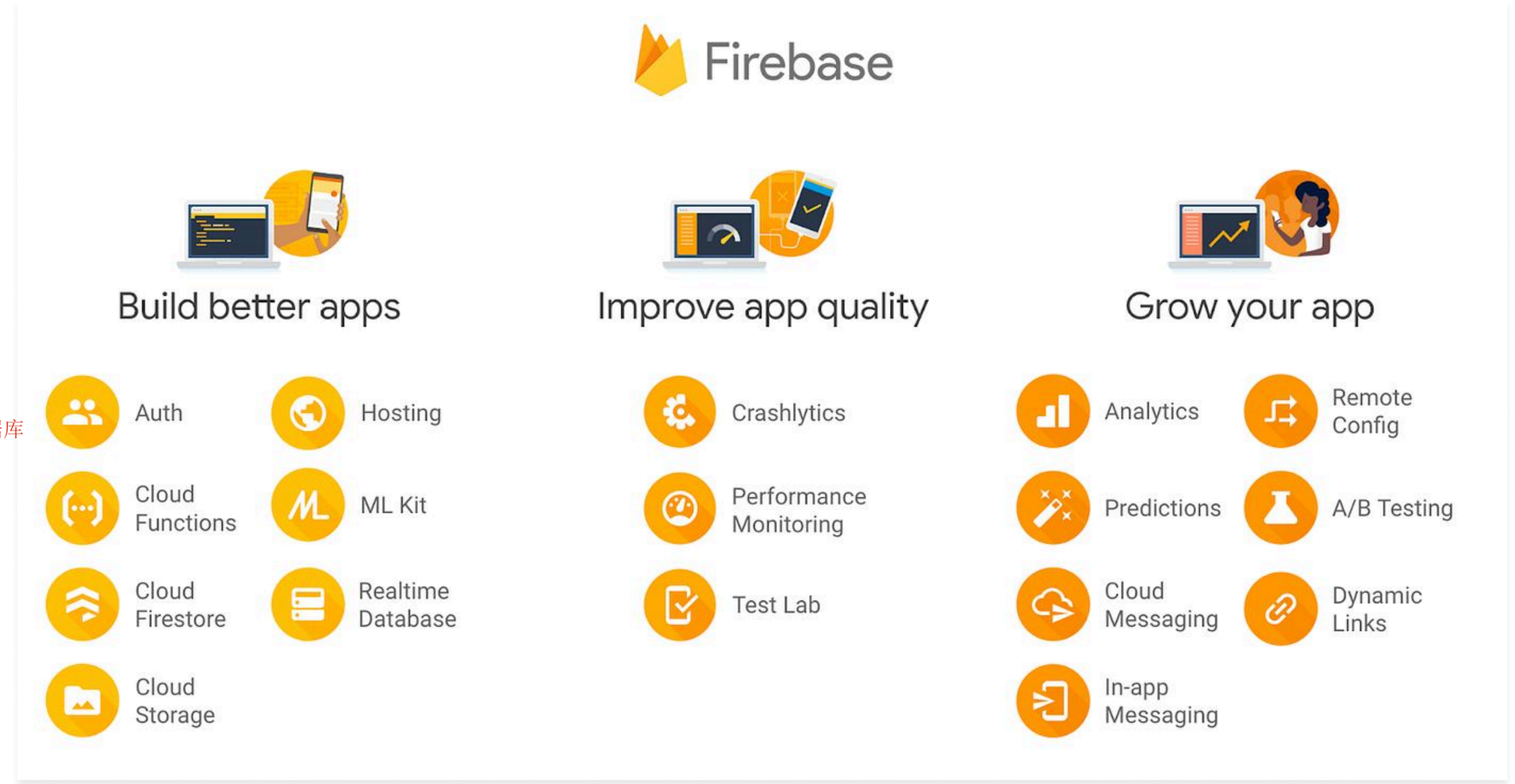


CRUD

例如保存一份地形参数是 Create，再打开它是 Read。

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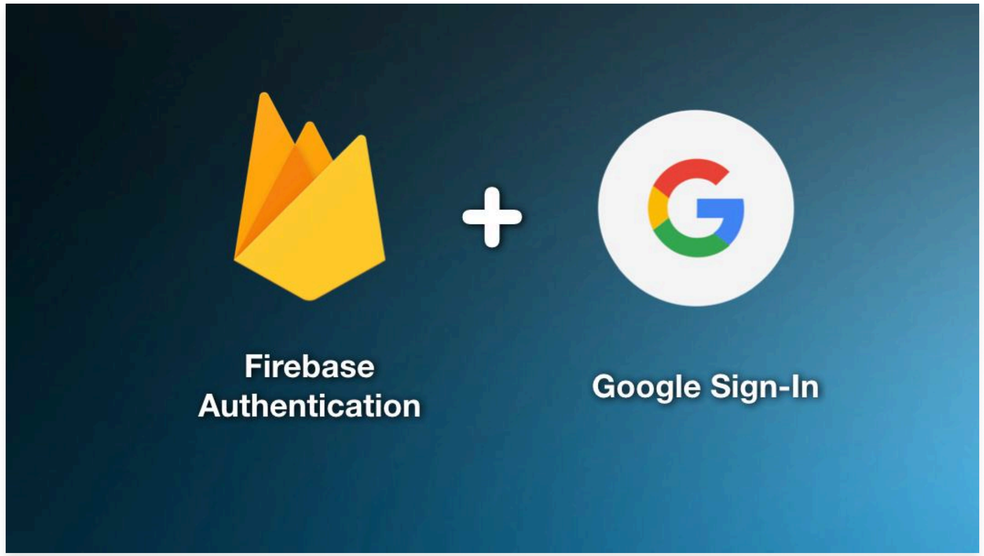
-Authetication 身份验证
-Firestore (Database) 数据库
-Storage 文件存储
-Hosting 网站托管



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Firebase

1 - Authetication



Firebase Authentication answers: "Who is using my application?"

负责知道“现在是谁登录”，并处理注册、登录和退出。

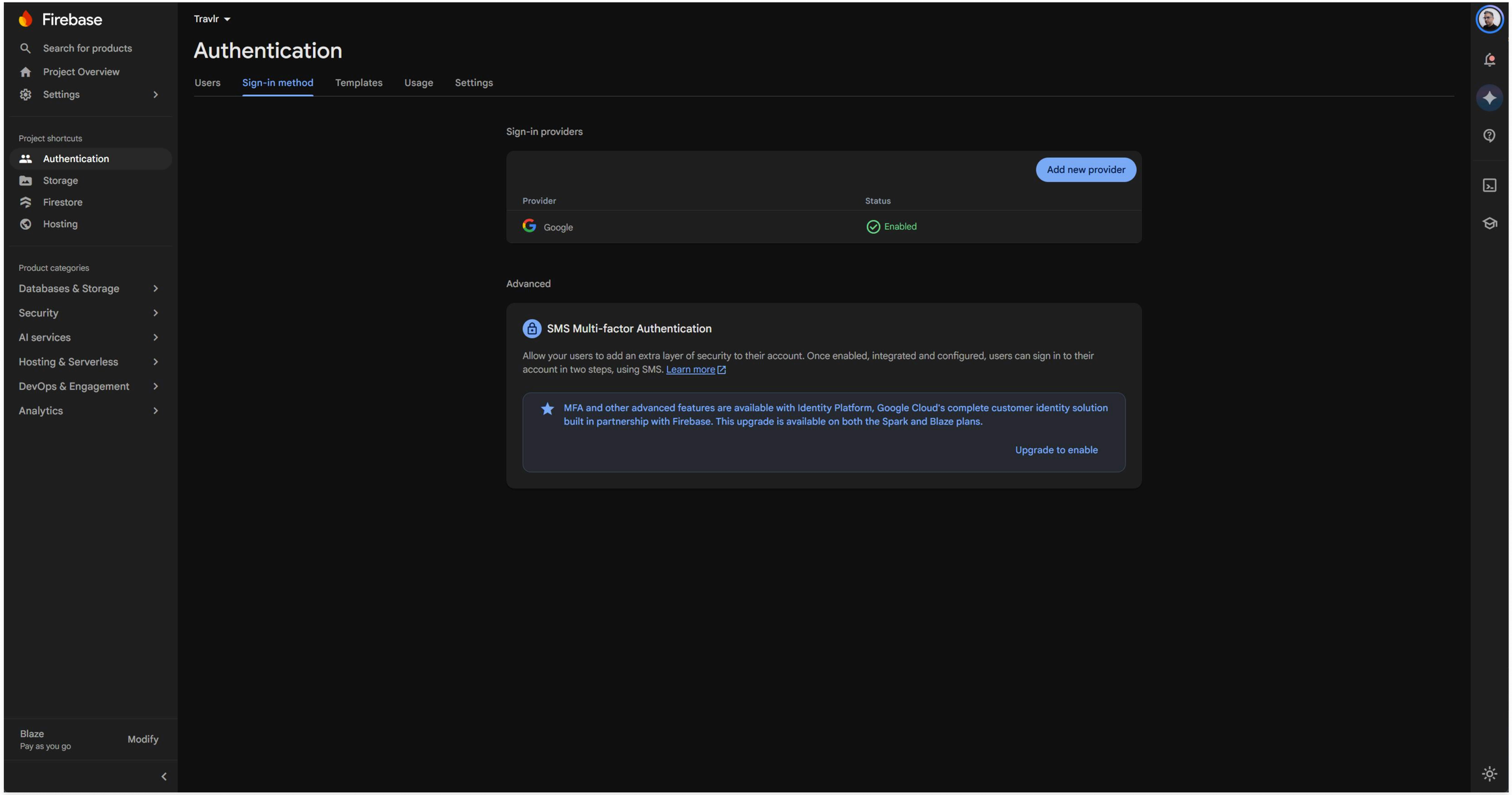
It manages user accounts and login/logout.

它本身不等于保存你的全部项目数据。

It can support email/password, Google, Apple, anonymous users, etc.

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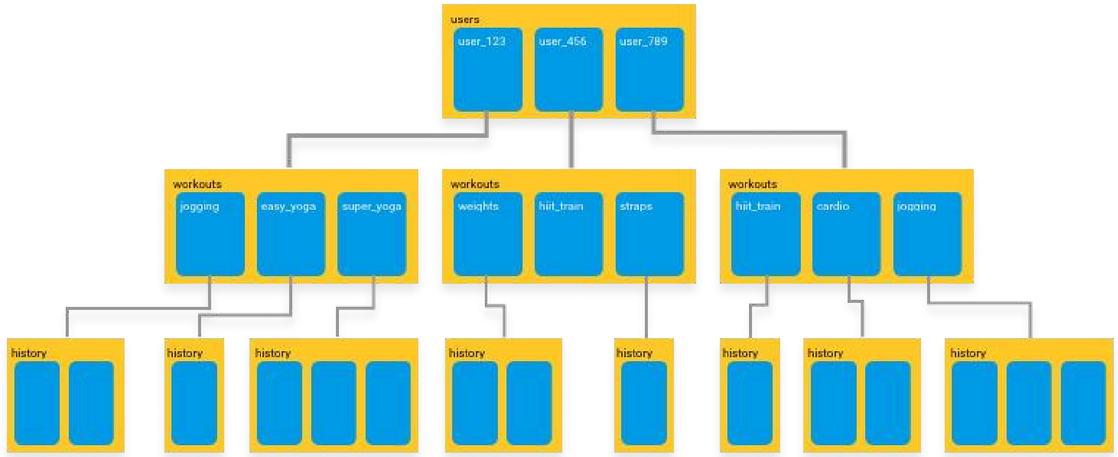
在 Firebase 项目中配置登录方式的位置。



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Firebase

2 - Firestore



Firestore stores structured information about your application.

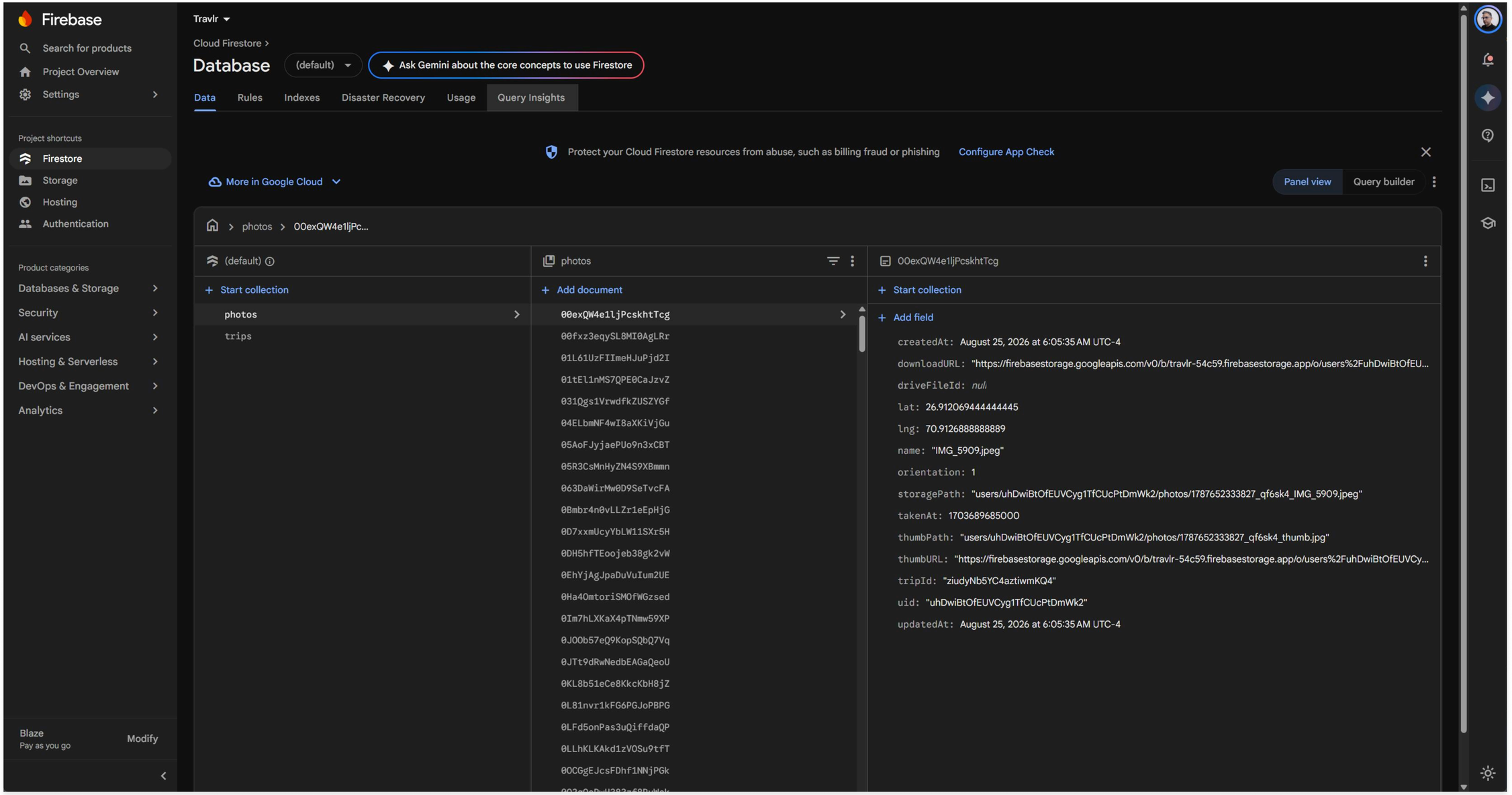
Think of it as the application's memory.

保存有结构的信息，例如用户 ID、某份地形配置的名称、噪声参数和创建时间。

可以把它理解为应用的“记录本”。

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展示数据库中的记录和字段。



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Firebase

3 - Storage

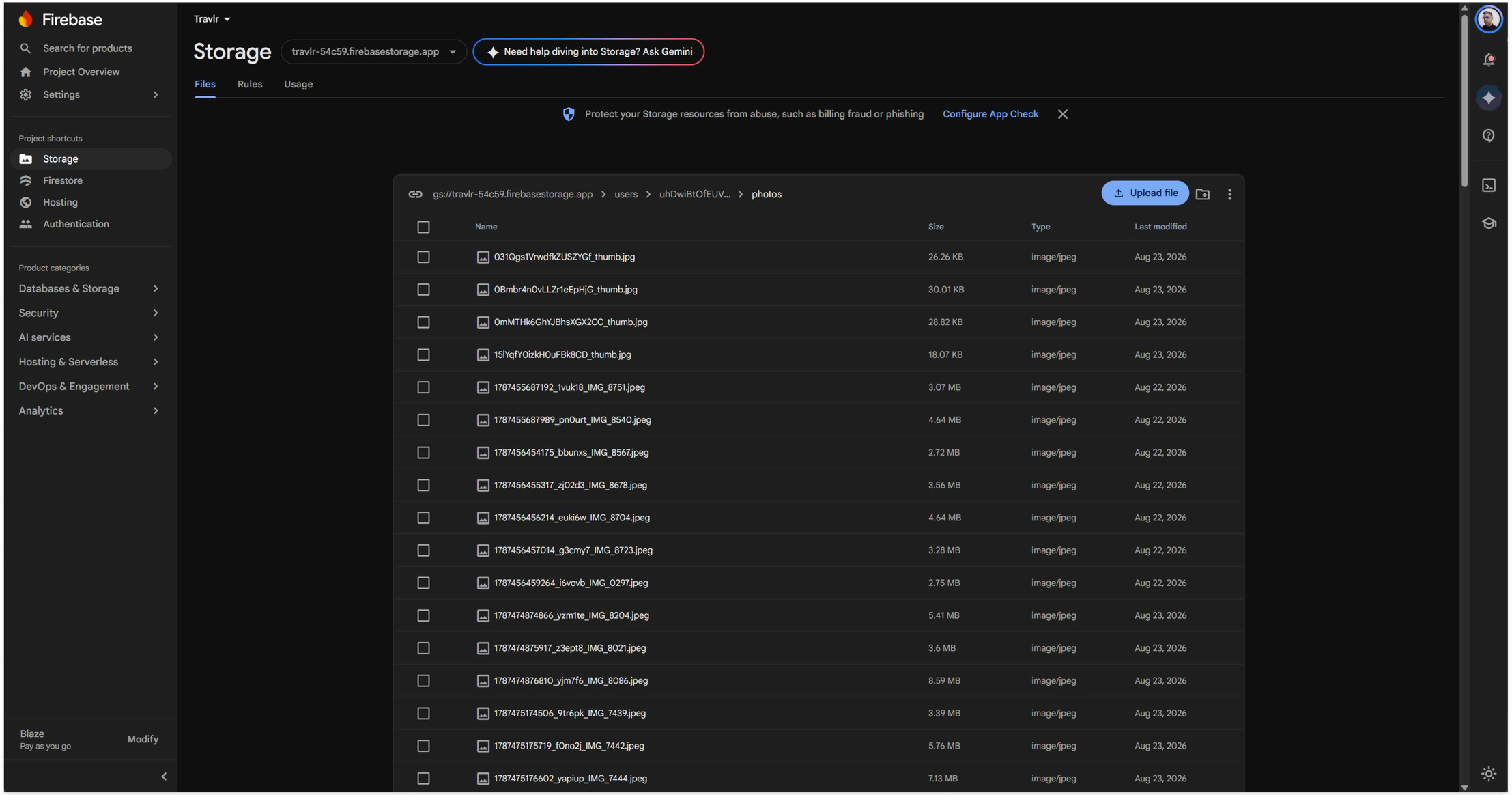


Firebase Storage stores the actual files used by your application.

适合放图片、视频、导出的模型等文件；与上一页的“参数和记录”分工不同。

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展示上传后的文件列表。



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Firebase

4 - Hosting

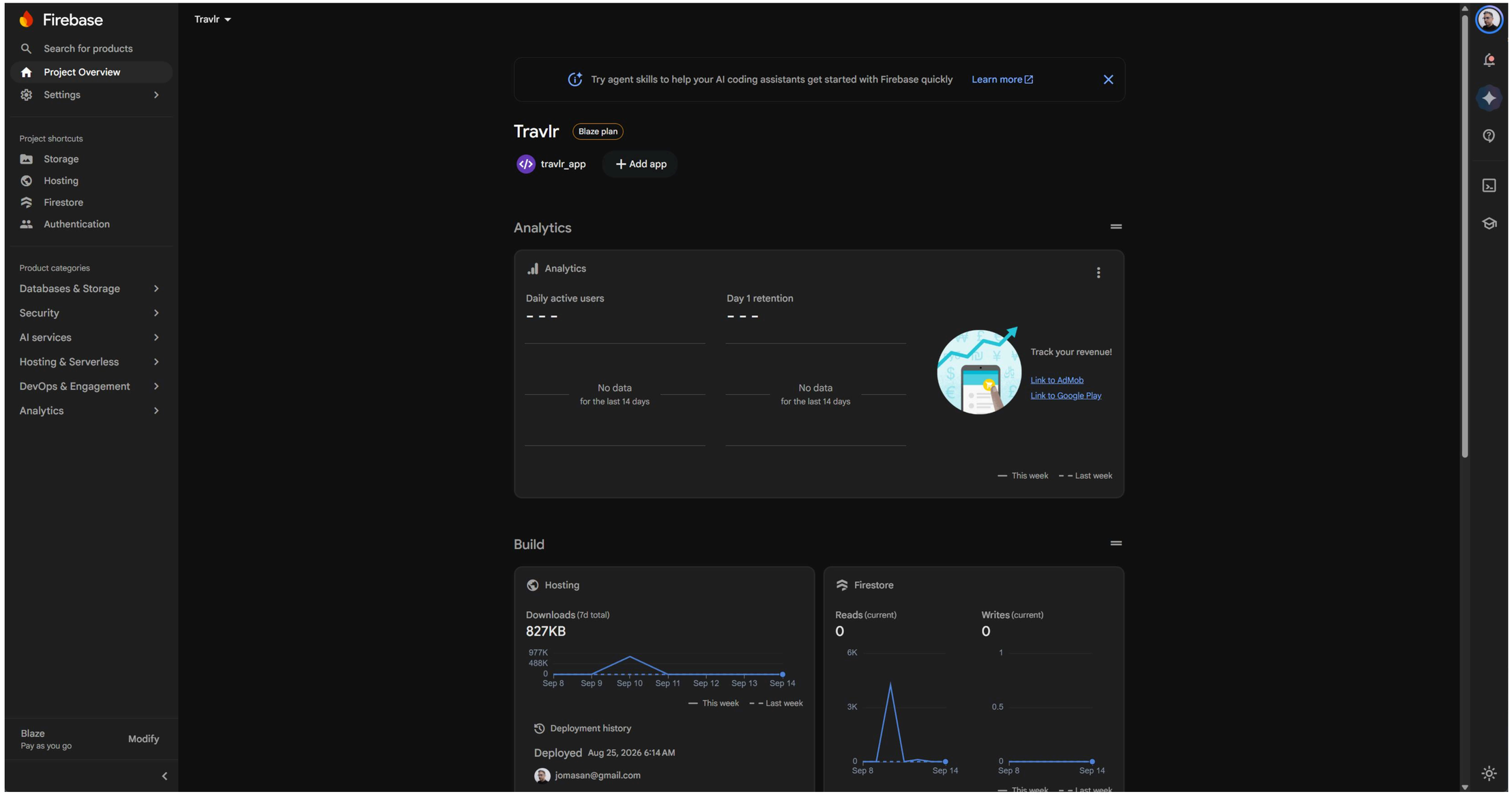


Firebase Hosting puts your web application on the internet.

把前端网站发布到网上，使别人能通过网址访问。

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Firebase 控制台示例。 展示项目在控制台中的管理界面，为后面的安全规则做铺垫。



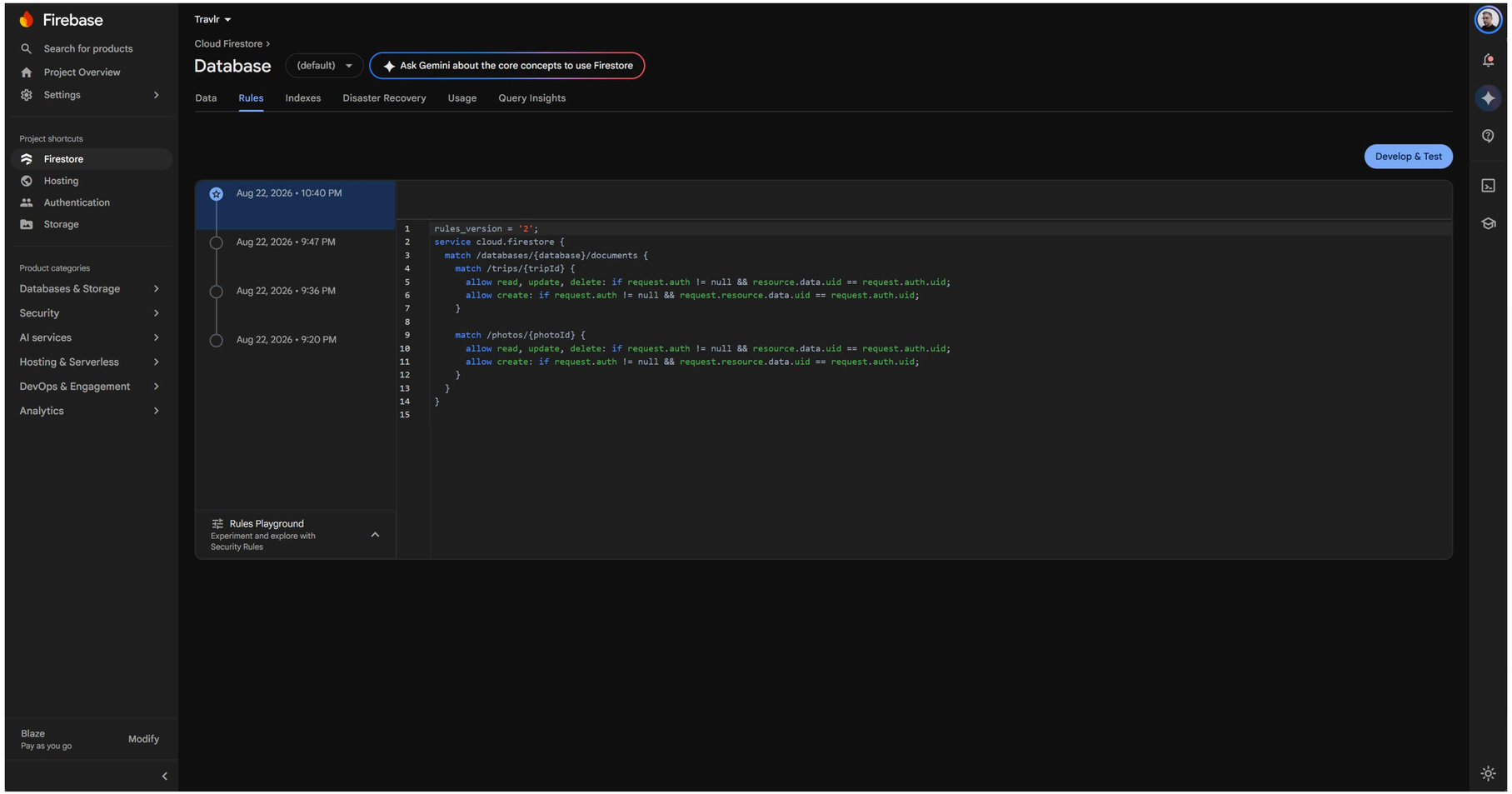
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Firebase

Security Rules: 安全规则

-Firestore Database

-Storage



Both Firestore and Storage need security rules.

These rules determine what authenticated—or unauthenticated—users are allowed to read, create, modify, or delete.

通过网络提供的软件服务。

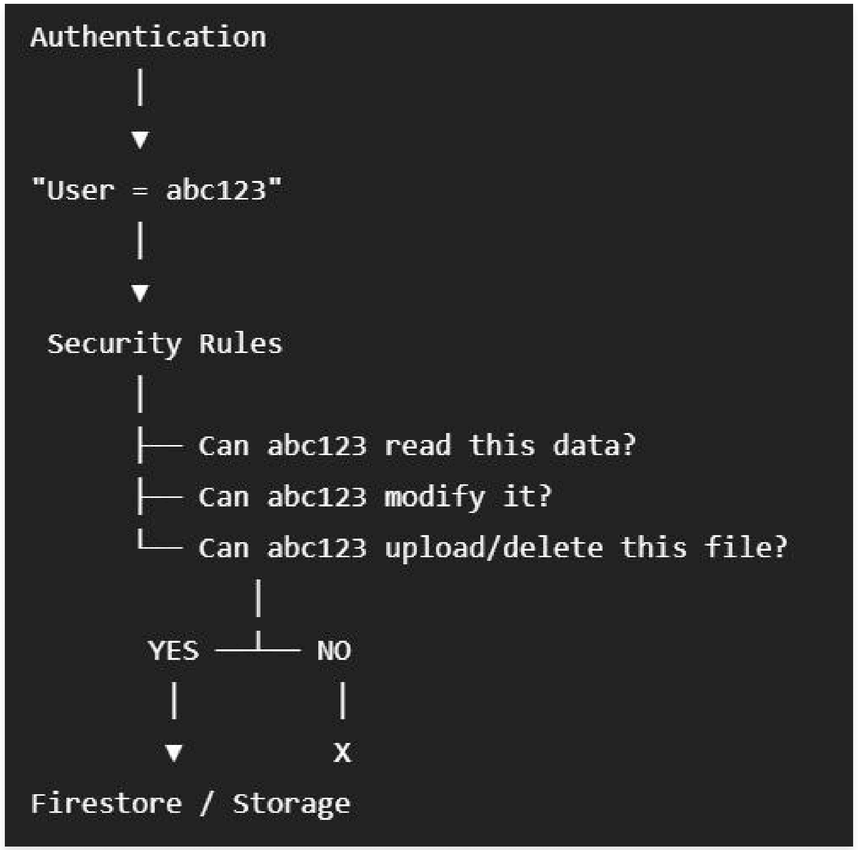
多个产品作例子，带出在线应用通常需要账户与持久化数据。
Firestore 和 Storage 都要定义谁可以读取、创建、修改、删除。用户能登录，不代表他就应该能读取其他用户的数据；具体权限由规则决定。

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规则的判断流程



Security Rules:
-Firestore Database
-Storage



先确认用户身份，

再检查这个用户对目标数据或文件有没有相应权限。

课件建议开发时把规则保存在代码仓库，便于管理修改记录。

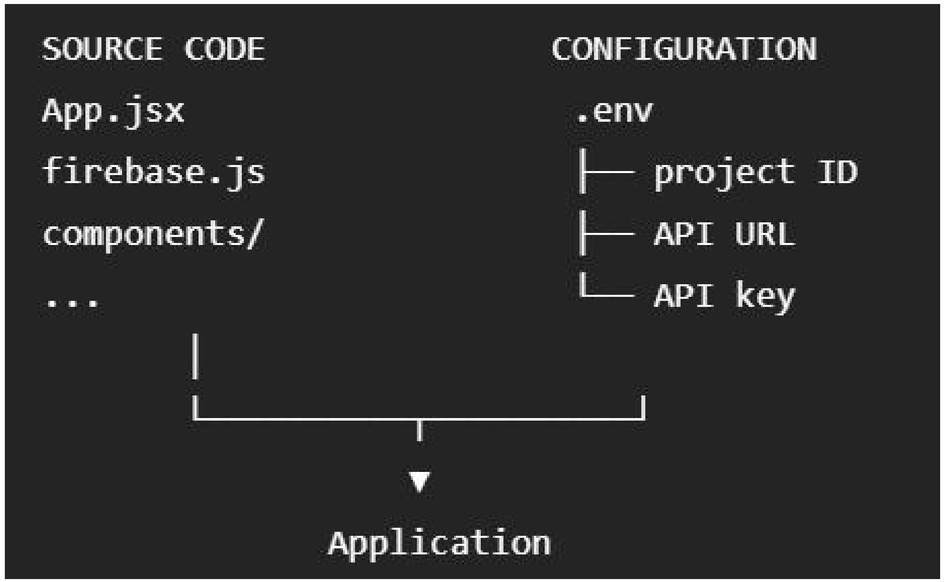
Rules can be edited manually in the Firebase Console, which is useful when initially configuring or testing a project.
However, for a development project it is generally better to keep the rules inside your code repository.

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Common Challenges:

- .env
- (environmental variables)



.env 环境变量文件。

把项目 ID 等配置集中放在一个地方，避免散落在代码各处。

A .env file stores values that your application needs but that you don't want hard-coded throughout the source code.

VITE_FIREBASE_API_KEY=...
VITE_FIREBASE_PROJECT_ID=my-project
VITE_FIREBASE_STORAGE_BUCKET=...

这里有个容易误会的点：前端 VITE_... 变量会进入浏览器端程序，不能把真正需要保密的服务端密钥放进去。

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Common Challenges:

- CORS:

CORS means Cross-Origin Resource Sharing.
Browsers restrict a website from freely requesting resources from a different origin.

浏览器访问另一个 origin（来源，通常由协议、域名和端口决定）的资源时，会受到跨来源规则限制。

这页是在提醒你：网站部署后，跨来源请求可能需要相应配置。

DESIGN 6197/4197 - Fall 2026
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自己写一份 Firebase 教程，并照着它启用 Authentication、Firestore、Storage、Hosting；
让应用保存与登录用户关联的数据。
最核心的实际功能是“保存并重新加载一份配置”，最后把公开网站链接填到指定表格。

Assignment 2:

-Create your own Firebase tutorial
-Follow your tutorial to enable authentication, firestore, storage and hosting
-Utilize your new functionality to store different data of your app linked to your login
-Main function to implement → Save and load a configuration
-Share your public website in the google spread sheet.